

namamk	ertemuanl topic	referensi	description
Arsitektur dan Organisasi Komputer	1 Introduction to Computer Organization & Architecture	Stalling W. 2016. Computer Organization & Architecture: Designing for Performance, 10th Ed.	Student are able to explain the main concept of Introduction to Computer Organization & Architecture
	Introduction to computer organization and architecture	William S. 2016. Computer Organization & Architecture	Students are able to explain the main concept of computer organization and architecture
	2 From silicone to chips	Stallings W. Computer Organization and Architecture: Designing for performance	Students are able to explain the main components inside the computer/ processors;
	Introduction to Computer Architecture	â€œComputer Organization and Architecture,â€ by William Stalling. Ninth Edition McGraw-Hill	Students are able to explain the process of making ke chip of processors.
	3 Inside the processor	Stallings W. Computer Organization and Architecture	Students are able to explain the main concept of computer architecture, especially Instruction Set Architecture (ISA)
	4 Electronic Circuit	Stallings W. Computer Organization and Architecture: Designing for performance	Students are able to explain the main concept of FEC and Pipeline inside the processor.
	Electronics and Circuit Design	â€œComputer Organization and Architecture,â€ by William Stalling. Ninth Edition McGraw-Hill	Students are able to explain the main concept of processors: pupeline
	5 CPU: Data path and control path	Computer Organization and Architecture,â€ by William Stalling. Ninth Edition McGraw-Hill,	Students are able to explain the main concept of electronic circuits that forms processors and its functionality.
		Stallings W., Computer Organization and Architecture: Designing for Performance	Students are able to explain the main components that construct the modern computer
	6 CPU		Students are able to explain the main concept inside the CPU: data and control path.
			Students are able to explain the main concept of data path and control inside the CPU.
			1) Students are able to explain the main function of CPU in the modern computer system.
			2) Students are able to explain the main concept of pipeline in the modern processor unit.
			3) Students are able to explain the main concept of Fetch-Execute Cycle in the modern processor unit.
Bahasa Inggris 2	CPU: Fetch and execute cycle and pipeline	Students are able to explain the main function of CPU in the modern computer system.	
	7 Review	Computer Organization and Architecture,â€ by William Stalling. Ninth Edition McGraw-Hill	Students are able to explain the main concept of pipeline in the modern processor unit.
	8 Presentasi dengan tema Multiprocessing System	â€œComputer Organization and Architecture,â€ by William Stalling. Ninth Edition McGraw-Hill.	Students are able to explain the main concept of Fetch-Execute Cycle in the modern processor unit.
	Workshop on assembly language	Stallings W. Computer Organization and Architecture: Designing for performance	Students are able to explain or to recap some of the important aspect in computer organization and computer architecture
	9 Workshop on MIPS MARS	Stallings, Computer Organization and Architecture	Students are able to explain the main concept behind processors, including the conceptual model, its components, FEC and its application
	workshop on mips mars4.5 #2	MIPS MARS Documentation	Students are able to explain the main application of multiprocessing system.
	10 Memory	â€œComputer Organization,â€ by Carl Hamacher, Zvonko Vranesic and Safwat Zaky. Fifth Edition McGraw-Hill, 2002.	Students are able to utilize instructions that available on MIPS32bit to solve problems
	Workshop on MIPS MARS	Charles W. Kann Introduction to MIPS assembly language	tudents are able to relate their understanding of ISA and FEC to the previous lecture.
	11 Memory and I/O	Stallings W. Computer Organization and Architecture	Students are able to create an assembly language as an implementation of ISA's.
	Performance & Instructions	Charles Kann, Introduction to MIPS assembly language	Students are able to code a program based on MIPS32 architecture
	12 Memory & Instruction Set Architecture (ISA)	William Stallings Designing Processor for performance	Students are able to explain the main concept of memory
	Memory and I/O for parallel computing	William Stallings, Designing processor for performance	Students are able to write MIPS program for solving problems
	13 Presentasi Multi Processors	Computer Organization and Architecture by William Stalling. Ninth Edition McGraw-Hill	students can explain the main concepts and its techniques
	Presentation on Multiprocessor	Stallings W. Computer Organization and Architecture	Students are able to explain the main concept of designing processor to fulfil the needs of
	14 Presentasi final project	Stallings, Computer Organization and Architecture	Students are able to explain the main function of Memory in the modern computer system.
Bahasa Inggris 2	1 listening	longman Toefl	Students are able to explain the main concept of ofÂ Instruction Set Architecture (ISA)
	2 Listening skills 3-6	longman Toefl	Students are able to explain the main role of memory and I/O in parallel computing
	3 Structure skills 1-3	longman Toefl	Students are able to present the concept of multi processors systems
	4 Structure skills 4-6	Longman-Toefl	Students are able to present their literature findings regarding to multiprocessor system
	5 Reading skills 4-6	Module A Toefl	Students are able to present the final project: literature study on super computer
	6 Reading Comprehension	longman Toefl	Students are able to present the results of the final project
	7 Listening skills 14-17	Module A Toefl	Syllabus, Listening skills 1-3
	8 Grammar and Written Expression	longman Toefl	listening skills 3-6
	9 listening	Longman-Toefl	Structure skills 1-3
	10 listening	Longman - TOEFL	Structure
	11 Reading skills 8-11	longman -toefl	Reading skills 4-6
	12 Grammar & written expression	Longman-Toefl	Reading Comprehension skill 1-3

Bahasa Inggris 2	13 Grammar and Written Expression	Longman - TOEFL	Grammar skill 33-37
	14 Review all skills on toefl	Toefl module A	Reading Toefl Test Simulation
Ethical Hacking	1 Later, the module discusses hacking and ethical hacking concepts an	CEH v.12 Module 1	Attackers break into systems for various reasons and purposes. Therefore, it is important to
	2 Chapter 2: Foot printing and Reconnaissance	CEH v12	This module starts with an introduction to foot printing concepts and provides insight into foot
	3 Chapter 3 Scanning Networks	CEH v12	This module starts with an overview of network scanning and provides an insight into various
	4 Chapter 4 Enumeration	CEH v13	The module provides an insight into different techniques for NETBIOS, SNMP, LDAP, NTP, SMTP, DNS,
	5 Chapter 05 Vulnerability	CEH v13	This module starts with an introduction to vulnerability assessment concepts. This module also
	6 Chapter 6 System Hacking	CEH v13	discusses the vulnerability assessment management life cycle and various approaches and tools used
	7 Chapter 6 System Hacking	CEEH v13	This module starts with an overview of hacking methodology. Later, it discusses in detail various
	8 Chapter 7 Malware	CEH v13	Malware is a program designed to perform malicious acts (The term itself is a contraction of
	9 Chapter 7 Malware	CEH v13	"malicious software"). Malware such a viruses, Trojans, worms, spyware, and rootkits allow an
	10 Chapter 8 Sniffing	CEH v13	Malware is commonly used by an attacker to compromise target systems. Preventing malware from
	11 Chapter 9 Social Engineering	CEH v13	This module begins with an overview of sniffing concepts and provides insight into MAC, DHCP, ARP,
	12 Vulnerability Assessment	NIST SP800-115	This module provides insight into human-based, computer-based, and mobile-based social
	13 Penetration Test	NIST 800-115	Understanding how to conduct Vulnerability Assessment
	14 Capita Selecta	CEHv13, NIST	Understanding the Security Professional's scope of work and security managed service industry.
Etika Profesi dan Hak Kekayaan Intelektual	1 Chapter 1	Buku Etika Profesi dan Hak Kekayaan Intelektual	Summarize all topics.
	2 Chapter 2	Pearson	Chapter 1
	3 Chapter 3	Mc Graw Hill	Chapter 2
	4 Chapter 4	Pearson	Chapter 3
	5 .	Pearson	Chapter 4
	6 Chapter 6	Pearson	. Chapter 6
	7 CHAPTER 7	Pearson	CHAPTER 7
	8 Chapter 8	Pearson	Chapter 8
	9 Chapter 9	Pearson	Chapter 9
	10 Chapter 10	Pearson	Chapter 10
	11 Chapter 11	Pearson	Chapter 11
	12 Pertemuan ke 12	Pearson	Chapter 12
	13 Chapter 13	Pearson	Chapter 13
	14 chapter 14	Pearson	Chapter 14
			INFORMATION SOCIETY :
	1 INFORMATION SOCIETY		- Definition
			- Characterstic Era Informasi
			- Dampak Positif
			- Dampak Negatif
			- Tantangan di Era Informasi
			- Case study
Informatika Sosial	introduction to social informatics	INFORMATION SOCIETY Baase,Sara. (2013). A Gift of Fire (4). England: Pearson Education.	Students are able to explain the main background why it is important to study social informatics.
	2 Introduction to Social Informatics Social Informatics: A New Persp		- Introduction
			- Socio-technical Interaction Networks: The Vitality of Electronic Journals
			- How Social Context Matters: Intranets in Action
			- How Work Process Matters
			- A Socio-technical Approach to ICT Infrastructures: Public Access to Information v
	the gift of fire	Article Journal Scopus Sara Baase, the gift of fire	students are able to explain the pros and cons of technology usages
	3 Ethical point of views	Baase S. The Gift of Fire	Students are able to explain some of the ethical point of view.
			- Social informatics menurut Rob Kling
			- History of social informatics
			- Social Networks
			- Social networks for business
	Social Informatics concepts & Social Networks	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future.	- The future social networks
		Cambridge Scholars	- Case study
	4 Privacy	Sara Baase, the gift of fire	Students are able to explain the main concept of privacy and its applications.

Informatika Sosial	4	Social and computational issues in Social Informatics: Social Informatics	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future. Cambridge Scholars	Social and computational issues in Social Informatics: Social Informatics & Privacy : - Social informatics and Privacy - Issues in Social - Issues in Computational - Case study
	5	Privacy and freedom of speech	Baase S. The Gift of Fire	Students are able to explain the main concept and its application of privacy and freedom of speech. Social and computational issues in Social Informatics: Freedom of Speech : - Introduction - Challenges and Opportunities of Freedom of Speech - How to resolve as a Government, as a Owner of platform, as a User media social account - Case study - Discussion
		Social and computational issues in Social Informatics: Freedom of Speech	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future. Cambridge Scholar	-
	6	Freedom of Speech & Intellectual Property Rights	Sara Baase, A Gift of Fire	Students are able to explain the consequences of the technology implementation in terms of intellectual properties such as copyright and patent. Students are able to present some of the example (case) of intellectual properties such as copyright and patent. Social and computational issues in Social Informatics: Intellectual Property : - Introduction - Challenges and Opportunities of Intellectual Property - Fair Usage Policy in Indonesia - Case study - Discussion
		Social and computational issues in Social Informatics: Intellectual Property	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future. Cambridge Scholars	-
	7	Review on ethics, privacy, fos dan ip	Sara Baase, the gift of fire	students are able to answers the questions regarding to ethics, privacy, fos dan ip Social and computational issues in Social Informatics: Freedom of Speech and Intellectual Property : - Introduction - Freedom of Speech - Intellectual Property - Case study
		Social and computational issues in Social Informatics: Freedom of Speech	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future. Cambridge Scholars	-
	8	Cyber space, Crime & work	A Gift of Fire (Fourth Edition); Sara Baase 2012 Pearson	Students are able to explain the consequences of the technology implementation in terms of ethical issues at work Social and computational issues in Social Informatics: Cyber Crime : - Introduction - Fraud - Fraud Diamond Theory - Financial Crime - Case Study
		Social and computational issues in Social Informatics: Cyber Crime	Baase S. 2012. A Gift of Fire (Fourth Edition) Pearson	
	9	Ruang siber	Sara Baase, the gift of fire	Mahasiswa dapat menjelaskan peran etika di dalam berinteraksi di ruang siber Social Informatics and computational issues in Social Informatics: Work Social and computational issues in Social Informatics: Risks Context - Introduction - Mitigation Risk in Social and Economy - Social Discrimination
		Social Informatics	Baase S. 2012. A Gift of Fire (Fourth Edition) Pearson	-
	10	Social and computational issues in Social Informatics: Risks Context	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future. Cambridge Scholars	-
		Work ethics	Sara Baase, the gift of fire	Students can explain the main principles of work ethics. Social and computational issues in Social Informatics: Failures Context - Introduction - Mitigation Failures, Root and Implication - Case study - Discussion
	11	Social and computational issues in Social Informatics: Failures Context	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future. Cambridge Scholars	-
		Work Ethics and Evaluating Technology	Sara Baase, the gift of fire	Students are able to conduct field research to gather data of how value, ethics and work ethics are

Informatika Sosial	12	Social and computational issues in Social Informatics: Evaluating a	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future. Cambridge Scholars	Social and Computational Issues in Social Informatics: Evaluating and Controlling Technology : - Definition - Evaluating Technology - Controlling Technology - Case Study - Discussion
	13	Work ethics Controlling the technology	Sara Baase, The Gift of Fire Baase S. The Gift of Fire	Students are able to present the research finding Students are able to explain the advantages and disadvantages of technology
		Social and computational issues in Social Informatics: Errors, Failu		Social and computational issues in Social Informatics: Errors, Failures, andn Risks : - Introduction - Errors - Failures - Risks - Case Study - Dissscussion
Interpersonal Skill	14	Professional ethics and responsibility	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future. Cambridge Scholars Sara Baase, the gift of fire	Students are able to explain the main concept of Professional ethics and responsibility in computing Social and computational issues in Social Informatics: Professional Ethics and Responsibilities : - Introduction - Primary Issues in Professional Ethics and Responsibilities - Computer Security - Computer Ethic for Professional
		Social and computational issues in Social Informatics: Professional	Rosenbaum H & Fichman P. 2014. Social Informatics: Past, Present and Future. Cambridge Scholars	- Case study - Discussion - Specific sub-topics: - Why interpersonal skill? - Understanding yourself
	1	INTRODUCTION INTERPERSONAL SKILL	John Hayes	Topic: What Predicets Success Specific sub-topics: 1. Introduction to the course 2. Why do some high potensial people fail and other succed? 3. Management skill: attitude, personality, and perception 4. Interpersonal Skill is skill in interacting with people or other party
		What Predicets Success Specific sub-topics:	Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	Specific sub-topics: 1. What is self-awareness? Benefit of self awareness 2. How to gain self-awareness? a. Self-analysis b. Behaviour c. Personality d. Attitudes e. Perceptions f. Attribution theory g. Other perceptions h. Self-disclosure i. Diverse experience
	2	Developing self awareness	Management skill for everyday life: the practical coach. Caproni, Paula J. 2004. Prentice Hall.	- Self-concept - Self-fulfilling
		Journey your self	Mc Shane & Glinow	

Interpersonal Skill	3	Self disclosure and Trust Specific	Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	1. What is Self disclosure? a. Benefit on self- disclosure b. Fears associated with Self disclosure c. The role of self disclosure in increasing 2. What is trust a. The role of trust in organizations b. Organizational trusts-builders c. A note of caution d. Bhari windows concept
		Self Disclosure: theory , Research & Therapy for interpersonal skill	Valerian J. Derlega, John H. Berg	-Private Self-Consciousness and Self-Disclosure Responsiveness and Self-Disclosure -The Relation of Loneliness and Self-Disclosure
	4	: Communicating Effectively Specific sub-topics:	Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International	1. Why do some high potential people fail and other succeed? 2. What is communication? Verbal and non-verbal 3. How to implement you-attitude while communicate a message? 4. Knowing your audience 5. Active listening 6. Assertive communication 7. Taking responsibility and message 8. Informative, positive, negative, and persuasion message
		CROSS CULTURE COMMUNICATION	Kitty O Locker	-Diversity and culture basics -Diversity on workplaces -High-context culture -Low-context culture -Dimension of culture
	5	Proposals and program report	Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	Topic: Proposals and program report Specific sub-topics: 1. What is report 2. What should I do before I write a proposal? 3. Structure of the report functions and characteristics of the report 4. Proposals for class research projects 5. Purpose of writing the research report 6. The types of proposals 7. The kinds of proposals 8. Terms of making proposals 9. The stage of preparation of proposals
		Understanding & Working with Diverse Others	Devito	-Perception -Emotional message -Conflict communication 1. What is culture? 2. How does culture affect business communication? 3. Consider diversity as part of audience analysis 4. Begin to be aware of the values, beliefs, and practices in other cultures 5. Become even more sensitive to verbal and nonverbal behavior
	6	Communicating Across Culture	Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	
		Etiquette and manner In the Workplace	Jane Carlson	-Etiquette in the Workplace provides step-by-step suggestions useful to all workers in their daily interactions with the public. -Overcome frustration at work by gaining confidence and taking control of common difficult

Interpersonal Skill

7	Buildings team and work groups	Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	- What is team work? - Why teams? - Potential limitation of teams - Types of teams - Team developmental stage - Characteristics of high performance teams - Tips for effective teams - Why team fail
	Guest Lecture Digital HR	All reference	FROM DATA TO DECISIONS: BUILDING A RESILIENT AND SECURE DIGITAL HR ECOSYSTEM - What is negotiation and Why is it important? Why do we negotiate - Benefit of honing negotiation skill - Integrative and distributive bargaining strategies - Five stage of negotiating - Strategies for negotiating effectively - Scripting - Framing - Managing - Additional Tipe for effectively negotiating - Special situations in negotiations
8	Negotiation	Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	Introduction to brand meaning - understand the meaning of the image - understand the meaning of reputation - how to explore self-image and develop it
	Personal Branding	Farco, S, R. 2019. The masterbook of personal branding. Quadrant.	-Influence Control Team -Mapping strategy role model extrovert and introvert leadership
9	Introvert Vs Extrovert Leadership	Swinton W Hudson	- What is conflict - Limited resources - Differences in goals/objective - Miscommunication - Differing attitudes, values and perceptions - Style difference - Why is conflict management important? - Sources of interpersonal conflict - Conflict management strategies - Conflict prevention techniques
	Managing conflict	Management skill for everyday life: the practical coach. Caproni, Paula J. 2004. Prentice Hall.	-Researching Jobs -Resumes -Job application letter -Job Interview -Follow "up" letter and job offers
10	Job hunting	Kitty O Locker	- What is decisions making - Why is decision making important? - The decition making process - What kind of decision maker are you? - Collaborative decision making
	Making decitions and solving problems creatively	Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	-Problem solving tool box: logic tree -Rock band and root causes -Potential root causes of the problem -Problem solving tool box: yes/no tree -Problem solving tool box: design plan
11	Problem solving	Kensuke Watanabe	

Interpersonal Skill	11	Problems solving creatively	What is problem solving?	Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	- Why is problem solving important? - Problem solving techniques - Brainstorming - Building consensus through multivoting - What is creatively and why is it important? - Strategies for increasing creativity in problem solving - Strategies for increasing individual creativity
	12	Emotion at work and emotional intelligent		Mc Shane Glinow	- Emotion at work - Kind of emotions - Emotion model influence behavior - Emotional intelligence - Bhari window
		Networking		Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	- What is networking? - The important and uses of networking - Developing an appropriate mindset for networking - Strategies for building an effective network
	13	Effective Teamwork		Mc Shane Glinow	- Choosing team member to create effective teamwork - Team design - Team size - Team characteristic - Setting team target
		Mentoring		Interpersonal skills in organization. De Janasz, Suzanne., Dowd, Karen O., Schneider, Beth Z. 2012. McGraw-Hill International.	- What is mentoring? - The importance of mentoring - Benefits of mentoring - Quality of an effective mentor - Types of mentoring relationship - Four stage of mentoring relationship - Limitation of mentoring
IT Project Management	14	Project Management		Management skill for everyday life: the practical coach. Caproni, Paula J. 2004. Prentice Hall.	- What is project management - Why important project management - Benefit of project management - Eight steps to managing projects - Project managements tolls
		Project management		9. Watanabe, K. 2009. Problem solving 101: a simple book for a smart people. Osaka: Diamond, Inc.	Final project management task Simulation and final project task Setelah membaca materi Bab 1 tentang Konsep Manajemen Proyek Teknologi Informasi (TI), mahasiswa diharapkan mampu: - Memahami konsep manajemen proyek TI - Mengidentifikasi aspek - aspek penting dalam manajemen proyek TI - Menjelaskan berbagai pendekatan dan atribut dalam manajemen proyek TI
	1	KONSEP MANAJEMEN PROYEK TEKNOLOGI INFORMASI (TI)		MANAJEMEN PROYEK TEKNOLOGI INFORMASI (TI) Dr. Ir. Ridwan Daud Mahande, S.Pd., M.Pd., IPM.	- Describe an overall framework for project integration management as it relates to the other project management knowledge areas and the project life cycle - Discuss the strategic planning process and apply different project selection methods - Explain the importance of creating a project charter to formally initiate projects - Describe project management plan development, understand the content of these plans, and Understand the growing need for better project management, especially for information technology (IT) projects
		Project Integration Management		Schwalbe K., 2016. Information Technology Project Management. 8th edition. Cengage USA.	Explain what a project is, provide examples of IT projects, list various attributes of projects, and describe the triple constraint of project management Describe project management and discuss key elements of the project management framework, including project stakeholders, the project management knowledge areas, common tools and techniques, and project success
	2	Introduction to Project Management		Schwalbe K., 2016. Information Technology Project Management. 8th edition. Cengage USA.	

IT Project Management	2	The Project Management and Information Technology Context	Information Technology Management Project Schwalbe 9th Edition	After reading this chapter, you will be able to: - â€ƒ Define the systems view of project management and how it applies to information technology (IT) projects - â€ƒ Summarize organizations, including the four frames, organizational structures, and organizational culture - â€ƒ Explain why stakeholder management and top management commitment are critical for a projectâ€™s success - â€ƒ Distinguish between project and product life cycles - â€ƒ Discuss the unique attributes and diverse nature of IT projects - â€ƒ Summarize recent trends affecting IT project management, including globalization, outsourcing, virtual teams, and agile project management
	3	The Project Management and Information Technology Context	Schwalbe K., 2016. Information Technology Project Management. 8th edition. Cengage USA. Chapter 2.	-Describe the systems view of project management and how it applies to information technology (IT) projects Le a r n i n g O b j e c t i v e s After reading this chapter, you will be able to: - â€ƒ Describe the five project management process groups, the typical level of activity for each, and the interactions among them - â€ƒ Relate the project management process groups to the project management knowledge areas - â€ƒ Discuss how organizations develop information technology (IT) project management methodologies to meet their needs - â€ƒ Review a case study of an organization applying the project management process groups to manage an IT project, describe outputs of each process group, and understand the contribution that effective initiating, planning, executing, monitoring and controlling, and closing make to project success - â€ƒ Review a case study of the same project managed with an agile focus and compare the key differences between an agile approach and a predictive approach
		The Project Management Process Groups	Information Technology Management Project. Schwalbe. 9th Edition	- â€ƒ Describe several templates for creating documents for each process group Describe an overall framework for project integration management as it relates to the other project management knowledge areas and the project life cycle - â€ƒ Discuss the strategic planning process and apply different project selection methods - â€ƒ Explain the importance of creating a project charter to formally initiate projects - â€ƒ Describe project management plan development, understand the content of these plans, and describe approaches for creating them
	4	Project Integration Management	Information Technology Management Project Schwalbe 9th Edition	- â€ƒ Describe the five project management process groups, the typical level of activity for each, and the interactions among them â€ƒ Understand how the project management process groups relate to the project management knowledge areas - â€ƒ Discuss how organizations develop information technology (IT) project management methodologies to meet their needs - â€ƒ Review a case study of an organization applying the project management process groups to
		The Project Management Process Groups: A Case Study	Schwalbe K., 2016. Information Technology Project Management. 8th edition. Cengage USA	List key reasons why good project scope management is important - â€ƒ Describe the process of planning scope management - â€ƒ Discuss methods for collecting and documenting requirements to meet stakeholder needs and expectations - â€ƒ Explain the scope definition process and describe the contents of a project scope statement - â€ƒ Discuss the process for creating a work breakdown structure using the analogy, top-down, bottom-up, and mind-mapping approaches - â€ƒ Explain the importance of validating scope and how it relates to defining and controlling scope - â€ƒ Given an information technology (IT) project situation, show how recommended approaches for controlling scope can improve the potential for project success
	5	Project Scope Management	Information Technology Management Project, Schwalbe, 9th Edition	

IT Project Management	5	Project Time Management	Schwalbe K., 2016. Information Technology Project Management. 8th edition. Cengage USA	- Understand the importance of project schedules and good project time management - Discuss the process of planning schedule management - Define activities as the basis for developing project schedules - Describe how project managers use network diagrams and dependencies to assist in activity sequencing - Discuss the process of planning schedule management - Define activities as the basis for developing project schedules - Describe how project managers use network diagrams and dependencies to assist in activity sequencing - Explain how various tools and techniques help project managers perform activity duration estimates - Use a Gantt chart for planning and tracking schedule information, find the critical path for a project, and describe how critical chain scheduling and the Program Evaluation and Review Technique (PERT) affect schedule development - Compare how schedule management is addressed using Agile vs. more predictive project approaches - Discuss how reality checks and discipline are involved in controlling and managing changes to the project schedule
	6	Project Schedule Management		- Understand the importance of good project scope management - Describe the process of planning scope management - Discuss methods for collecting and documenting requirements to meet stakeholder needs and expectations - Explain the scope definition process and describe the contents of a project scope statement - Explain the importance of validating scope and how it relates to defining and controlling scope - Understand the importance of controlling scope and approaches for preventing scope-related problems on information technology (IT) projects
		Project Scope Management	Information Technology Management Project Schwalbe, 9th Edition	- Describe how software can assist in project scope management - Develop a justification for project cost management and its importance in achieving project success - Explain basic project cost management principles, concepts, and terms - Describe the process of planning cost management - Discuss different types of cost estimates and methods for preparing them - Using an example of an information technology (IT) project, list and describe the processes of determining a budget and preparing a cost estimate - Justify the use of earned value management and project portfolio management to assist in cost control - Describe how project management software can assist in project cost management - Discuss considerations for agile/adaptive environments
	7	Project Cost Management	Schwalber, Kathy, "Information Technology Project Management".	- Understand the importance of project cost management - Explain basic project cost management principles, concepts, and terms - Describe the process of planning cost management - Discuss different types of cost estimates and methods for preparing them - Understand the processes of determining a budget and preparing a cost estimate for an information technology (IT) project
			Information Technology Management Project Schwalbe 9th Edition	- Understand the benefits of earned value management and project portfolio management to assist in achieving project success for information technology (IT) products and services - Define project quality management and understand how quality relates to various aspects of IT projects - Describe quality management planning and how quality and scope management are related - Discuss the importance of managing quality and quality assurance - Explain the main outputs of the quality control process
	8	Presentation	Schwalbe, Kathy, "Information Technology Project Management". Presentation	
		Project Quality Management	Information Technology Management Project Schwalbe 9th Edition	

IT Project Management	9	Project Quality Management	Schwalbe, Kathy, "Information Technology Project Management".	-Understand the importance of project quality management for information technology (IT) products and services -Define project quality management and understand how quality relates to various aspects of IT projects -Describe quality management planning and how quality and scope management are related -Discuss the importance of quality assurance -Explain the main outputs of the quality control process -Understand the tools and techniques for quality control, such as the Seven Basic Tools of Quality, statistical sampling, Six Sigma, and testing -Summarize the contributions of noteworthy quality experts to modern quality management
		Project Quality Management	Information Technology Management Project Schwalbe, 9th Edition	People are the most important assets in organizations and on projects. Therefore, it is essential for project managers to be good human resource managers. They must also make effective use of physical resources. The major processes of project resource management include planning resources, Discuss the role of soft skills in IT project management, and highlight the importance of good communications as one means of achieving project success
	10	Project Communications Management	Information Technology Management Project. Schwalbe 9th Edition	- Review key concepts related to communications - Explain the elements of planning project communications and how to create a communications management plan - Describe how to manage communications, including communication technologies, media, and performance reporting - Discuss methods for controlling communications to ensure that information needs are met throughout the life of the project
		Project Human Resource Management	Schwalbe, Kathy, "Information Technology Project Management".	- Explain the importance of good human resource management on projects, including the current state of the global IT workforce and future implications for it - Define project human resource management and understand its processes - Summarize key concepts for managing people by understanding theories of motivation, - Understand the importance of good communications on projects and the need to develop soft skills, especially for IT project managers and their teams - Review key concepts related to communications
	11	Project Communications Management	Schwalbe, Kathy, "Information Technology Project Management".	- Explain the elements of planning project communications and how to create a communications management plan - Describe how to manage communications, including communication technologies, media, and performance reporting
		Project Risk Management	Information Technology Management Project, Schwalbe 9th Edition	- Discuss the elements of planning risk management and the contents of a risk management plan - List common sources of risks on information technology (IT) projects - Describe the process of identifying risks and create a risk register and risk report - Discuss qualitative risk analysis and explain how to calculate risk factors, create probability/impact matrixes, and apply the Top Ten Risk Item Tracking technique to rank risks
	12	Project Procurement Management	Information Technology Management Project Schwalbe, 9th Edition	- Describe the work involved in planning procurements for projects, including determining the proper type of contract to use and preparing a procurement management plan, statement of work, source selection criteria, and make-or-buy analysis - Discuss how to conduct procurements and strategies for obtaining seller responses, selecting sellers, and awarding contracts - Describe the process of controlling procurements by managing procurement relationships and monitoring contract performance - Discuss types of software that are available to assist in project procurement management
			Schwalbe, Kathy, "Information Technology Project Management".	- Understand the importance of project procurement management and the increasing use of outsourcing for information technology (IT) projects - Describe the work involved in planning procurements for projects, including determining the proper type of contract to use and preparing a procurement management plan, statement of work, source selection criteria, and make or buy analysis - Discuss how to conduct procurements and strategies for obtaining seller responses , selecting

IT Project Management	13	Project Risk Management		Understand risk and the importance of good project risk management Discuss the elements of planning risk management and the contents of a risk management plan List common sources of risks on information technology (IT) projects Describe the process of identifying risks and create a risk register Discuss qualitative risk analysis and explain how to calculate risk factors, create probability/impact matrixes, and apply the Top Ten Risk Item Tracking technique to rank risks Explain quantitative risk analysis and how to apply decision trees, simulation, and sensitivity analysis to quantify risks Provide examples of using different risk response planning strategies to address both negative and positive risks Discuss how to control risks Students are able to explain the main objective of Project Management and explain some of the related examples regarding to IT Project Management Mahasiswa dapat mempresentasikan hasil akhir proyek IT Understand the importance of project stakeholder management throughout the life of a project Discuss the process of identifying stakeholders, how to create a stakeholder register, and how to perform a stakeholder analysis Describe the contents of a stakeholder management plan Understand the process of managing stakeholder engagement and how to use an issue log effectively Formation of differential equation. Solution to First Order Differential Equations. Solutions to 1st order DE: - Direct integration. - Variable separation. - Homogenous equations. - Integrating factor. 2nd DE. Homogenous solution. Non-homogeneous solution. Methods of undetermined coefficients. Methods of Variations of Parameters. Introduction. Euler's method. Improved Euler's method. Infinite sequence. Infinite series. - Infinite Series - The Divergence and Integral Tests - Comparison Tests - Alternating Series - Ratio and Root Tests Review midtest. Parametric Equations. Calculus of Parametric Curves. Polar Coordinates. Area and Arc Length in Polar Coordinates. Vectors in the Plane. Vectors in Three Dimensions. The Dot Product. The Cross Product. Equations of Lines and Planes in Space. Calculus of parametric curves. Derivative of parametric equations Second-order derivative of parametric equations. Area under a parametric curve. Arc length of a parametric curve. Surface area generated by a parametric curve.
		Review and Study Case of Project Management	Schwalbe, Kathy, "Information Technology Project Management". James Cadle and Donald Yeates, 2008, Project Management for Information System, 5th Ed	
	14	Presentasi Proyek Akhir Mata Kuliah Manajemen Proyek IT	Kathy Schwalbe, Information Technology Project Management, 8Eds	
Kalkulus II		Project Stakeholder Management		
			Schwalbe, Kathy, "Information Technology Project Management".	
	1	Introduction, First Order Differential Equation	Ch 9	
	2	First-order Differential Equations.		
			Textbook, Slides.	
	3	2nd Order Differential Equation.		
			Textbook	
	4	Non-homogenous 2nd order differential equation.	Textbook Ch. 17	
	5	Euler method for 1st order ODE.		
			Textbook, Notes, Colab	
	6	Infinite Sequence and Series	Textbook Ch. 11	
	7	Infinite Series		
			Textbook.	
	8	Review midtest	Textbook, notes, colab.	
	9	Parametric Equation		
			Textbook, Openstax Ch. 7, Notes, Colab.	
	10	Vectors.		
			Slides, notes, textbook.	
	11	Parametric Equations (Continued)		
			Slide, Notes, Textbook.	

Kalkulus II	12	Vector, vector functions.	Slide, Notes, Textbook.	Dot product. Cross product. Triple scalar. Vector function. Calculus of vector valued functions.
	13	Vector-valued functions.	Slides, notes, textbook (Openstax).	Arch length. Curvature.
	14	Review	Slide, textbook	Quiz Introduction to wireless comms. Signals fundamentals. Evolutions of celluler networks. Transmission fundamentals. Transmission Fundamentals
Komunikasi Nirkabel	1	Introduction to wireless communications. Transmission Fundamentals	Stallings Ch. 1 Stalling, W., "Wireless Communications & Networking" Prentice Hall, 2nd Ed, 2005	Network topology. Switching Techniques. Circuit switching vs Packet Switching. ATM . Communication Networks
	2	Communication Network Communication Networks	Textbook Ch. 2 Stallings, W., "Wireless Communication and Networking".	Timing diagram of communication network. TCP/IP. Protocols and the TCP/IP Suite
	3	Communication Network and TCP/IP Protocols and the TCP/IP Suite	Texbook (Stallings) Ch. 2 & 3 Stallings, William, "Wireless Communication and Networking".	Antennas. Propagation model. LoS transmission. Fading. Antennas and Propagation
	4	Antenna and Propagation. Antennas and Propagation	Textbook, Ch. 5 Stallings, William, "Wireless Communication and Networking".	- Free Space Loss - Antenna Gain - Power budget in wireless transmission link Signal Encoding Techniques Coding and Error Control
	5	FSL, antenna gain, transmitted/received power Signal Encoding Techniques	Textbook Ch. 5 Stallings, William, "Wireless Communication Networks".	- Encoding vs Modulation. - Digital Modulations. - Analogue Modulations - Amplitude Modulation (AM) - Phase Modulation (PM) - Frequency Modulation (FM) Spread Spectrum
	6	Coding and Error Control Signal Encoding Techniques (1)	Stallings, William, "Wireless Communication and Networking". Textbook Ch. 6 - Part 1-2	Cellular Wireless Networks
	7	Analog Modulation Spread Spectrum	Textbook Ch. 6 Stallings, William, "Wireless Communication and Networking".	- FHSS. - DSSS - CDMA Review midtest Satellite Communications - Parity Check - CRC - Modulo 2 arithmetic - Polynomial code arithmetic Mobile IP and Wireless Application Protocol
	8	Cellular Wireless Networks Spread Spectrum	Stallings, William, "Wireless Communication and Network". Textbook, Notes.	Bluetooth Techniques Block coding. Hamming codes.
	9	Review midtest Satellite Communications	Slide, notes Stallings, William, "Wireless Communication and Networking".	
	10	Coding and Error Control Mobile IP and Wireless Application Protocol	Textbook and slides. Stallings, William, "Wireless Communication and Networking". Stallings, William, "Wireless Communication and Networking".	
	11	Bluetooth Techniques Linear Block code	Notes, Slides.	

Komunikasi Nirkabel	12	Convolutional Coding	Slide, textbook, notes.	Basic convolutional coding.
		Cordless Systems and Wireless Local Loop	Stallings, William, "Wireless Communication and Networking".	State diagram.
	13	Maximum Likelihood Decoding	Slides, notes, textbook.	Trellis diagram.
		Wireless LAN Technology	Stallings, William, "Wireless Communication and Networking".	Using trellis diagram for encoding.
Konsep Sistem Informasi	14	IEEE 802.11 Wireless LAN Standard	Stallings, William, "Wireless Communication and Networking".	Cordless Systems and Wireless Local Loop
		Wireless Cellular Network		Viterbi Decoding.
			Slide, notes, video presentation.	Maximum Likelihood Decoding.
	1	Week#1 - Pengenalan Sistem Informasi	Stair, Ralph & Reynolds, George. 2018. Principle of Information System. Cengage Learning. Thirteenth Edition.	Examples.
	2	Week#2 - Sistem Informasi dalam Organisasi	Ralph M. Stair and George W. Reynolds (2018). Principle of Information Systems, 13Edition. Cengage Learning.	Wireless LAN Technology
	3	Week#3 - Hardware dan Mobile Devices	Ralph M. Stair and George W. Reynolds (2018). Principle of Information Systems, 13Edition. Cengage Learning.	IEEE 802.11 Wireless LAN Standard
	4			Introduction to cellular network.
	5	Week#5 - Software dan Mobile Application	Ralph M. Stair and George W. Reynolds (2018). Principle of Information Systems, 13Edition. Cengage Learning.	History of wireless cellular network.
	6	Week#6 - Sistem Basis Data dan BIG Data	Topic: Database and Big Data	Evolution of cellular network.
			Specific sub-topics:	Towards 5G and 6G cellular network.
			1. Data Fundamental	1. Definisi Sistem Informasi
			2. Data Modeling and Database Characteristic	2. Data vs Informasi
			3. RDBMS	3. Sistem Informasi Bisnis
			4. Big Data	4. Konsep Teknologi Informasi
			5. Technology Used to Process Big Data	5. Sistem Informasi dalam Bisnis dan Masyarakat
	7	Week#7 - Presentasi	-	1. Definisi Organisasi
	8	Week#8 - Presentasi	-	2. Perubahan Organisasi
	9	Week#9 - Networks dan Cloud Computing	Ralph M. Stair and George W. Reynolds (2018). Principle of Information Systems, 13Edition. Cengage Learning.	3. Inovasi Sistem Informasi
	10	Guest Lecture Mindshare Meetup "Membangun SDM Unggul melala	-	4. Budaya Organisasi
				5. Karir dalam Sistem Informasi
				Membahas mengenai:
				1. Pendahuluan
				2. Perangkat Keras Komputer
				3. Anatomi Komputer
				4. Input dan Output Devices
				5. Jenis Komputer
				1. Pendahuluan
				2. An Overview of Software
				3. Komponen Software
				4. Ruang Lingkup Software
				5. Latihan
				1. Data Fundamentals
				2. Data Modeling dan Karakteristik Database
				3. Pentingnya DBMS
				4. Pengantar BIG Data
				5. Sumber BIG Data
				6. Penggunaan BIG Data
				7. Tantangan BIG Data
				Presentasi materi yang telah dibahas untuk masing-masing kelompok
				Tiap Kelompok lakukan presentasi sesuai materi yang telah dipilih
				1. Networks Fundamental
				2. The Internet dan Web
				3. Internet of Things
				4. Cloud Computing
				Mengikuti Guest Lecture Prodi SIF x LPkM

Konsep Sistem Informasi	11	Week#12 - Enterprise Resource Planning (ERP)	Ralph M. Stair and George W. Reynolds (2018). Principle of Information Systems, 13Edition. Cengage Learning. Any source from Google for Picture	1. Transaction Processing System 2. Jenis-Jenis TPS 3. Enterprise System
	12	Week#10 - E-Commerce dan M-Commerce	Ralph M. Stair and George W. Reynolds (2018). Principle of Information Systems, 13Edition. Cengage Learning. Any source from Google for Picture	1. Pengantar E-Commerce 2. Pengantar M-Commerce 3. E-Government 4. Aplikasi e-Commerce dan m-Commerce 5. Strategi Kesuksesan e-Commerce dan m-Commerce 6. Infrastruktur Teknologi
	13	Week#13 - Latihan Soal dan Quiz	-	1. Quiz Materi ERP 2. Latihan Soal Pra-Final Exam
	14	Week#14 - Guest Lecture "Knowledge and Specialized Management	-	Kuliah tamu interaktif dengan dosen Universitas Negeri Makassar 1. Introduction to Networking Today 2. Network Affect our Lives 3. Network Components 4. Network Representation and Topologies 5. Common Types of Networks 6. Internet Connections 7. Reliable Networks 8. Network Trends 9. Network Security 10. The IT Professional
LAN and Wireless	1	Networking Today & Basic Switch and End Device Configuration	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=enUS&deliveryType=INSTRUCTOR_LED (Chapter 1, 2)	11. Introduction of Basic Switch and End Device Configuration 12. Cisco IOS Access 13. IOS Navigation 1. Introduction of Protocols and Models 2. The Rules 3. Protocols 4. Protocol Suite 5. Standards Organizations 6. Reference Models 7. Data Encapsulation 8. Data Access 9. Introduction to Physical Layer 10. Purpose of the Physical Layer 11. Physical Layer Characteristics 12. Copper Cabling 13. UTP Cabling 14. Fiber-Optic Cabling 15. Wireless Media
	2	Protocols and Models & Physical Layer	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=enUS&deliveryType=INSTRUCTOR_LED	1. Introduction to Number Systems 2. Binary Number System 3. Hexadecimal Number System 4. Introduction to Data Link Layer 5. Purpose of the Data Link Layer 6. Topologies 7. Data Link Frame 1. Introduction to Ethernet Switching 2. Ethernet Frame 3. Ethernet MAC Address 4. The MAC Address Table 5. Switch Speeds and Forwarding Methods
	3	Number Systems and Data Link Layer	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=enUS&deliveryType=INSTRUCTOR_LED (Chapter 5, 6)	1. Introduction to Network Layer 2. Network Layer Characteristics 3. IPv4 Packet 4. IPv6 Packet 5. How a Host Routes 6. Router Routing Tables
	4	Ethernet Switching	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=en-US&deliveryType=INSTRUCTOR_LED (Chapter 7)	
	5	Network Layer	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=en-US&deliveryType=INSTRUCTOR_LED (Chapter 8)	

LAN and Wireless

6	Address Resolution	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=en-US&deliveryType=INSTRUCTOR_LED (Chapter 9)	1. Introduction to Address Resolution 2. MAC and IP 3. ARP 4. IPv6 Neighbour Discovery
7	Basic Router Configuration and Quiz before Mid Term Examination	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=en-US&deliveryType=INSTRUCTOR_LED (Chapter 10)	1. Introduction to Basic Router Configuration 2. Configure Initial Router Settings 3. Configure Interfaces 4. Configure the Default Gateway 1. Introduction to IPv4 Addressing 2. IPv4 Address Structure 3. IPv4 Unicast, Broadcast and Multicast 4. Types of IPv4 Addresses 5. Network Segmentation 6. Subnet an IPv4 Network 7. Subnetting a /16 and a /8 Prefix 8. Subnetting to Meet Requirements 9. Variable Length Subnet Mask (VLSM) 10. Structured Design 1. Introduction to IPv6 Addressing 2. IPv4 Issues 3. IPv6 Address Representation 4. IPv6 Address Types 5. GUA and LLA Static Configuration 6. Dynamic Addressing for IPv6 GUAs 7. Dynamic Addressing for IPv6 LLAs 8. IPv6 Multicast Addresses
8	IPv4 Addressing	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=enUS&deliveryType=INSTRUCTOR_LED (Chapter 11)	
9	IPv6 Addressing	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=en-US&deliveryType=INSTRUCTOR_LED (Chapter 12)	
10	ICMP	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=enUS&deliveryType=INSTRUCTOR_LED (Chapter 13)	1. Introduction to ICMP 2. ICMP Messages 3. Ping and Traceroute Tests
11	Transport Layer	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=en-US&deliveryType=INSTRUCTOR_LED (Chapter 14)	1. Introduction to Transport Layer 2. Transportation of Data 3. TCP Overview 4. UDP Overview 5. Port Numbers 6. TCP Communication Process 7. Reliability and Flow Control 8. UDP Communication
12	Application Layer	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=enUS&deliveryType=INSTRUCTOR_LED (Chapter 15)	1. Introduction to application layer 2. Application, Presentation and Session 3. Peer-to-Peer 4. Web and Email Protocols 5. IP Addressing Services 6. File Sharing Services
13	Network Security Fundamentals	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=en-US&deliveryType=INSTRUCTOR_LED (Chapter 16)	1. Introduction to Network Security Fundamentals 2. Security Threats and Vulnerabilities 3. Network Attacks 4. Network Attacks Mitigation 5. Device Security
14	Build a Small Networks	CCNA version 7.02, Module 1, Introduction to Networks v.7.02 (ITN) https://www.netacad.com/courses/ccna-introduction-networks?courseLang=en-S&deliveryType=INSTRUCTOR_LED (Chapter 17, 1-17)	1. Introduction to Build a Small Network 2. Devices in a Small Network 3. Small Network Applications and Protocols 4. Scale to Larger Networks 5. Verify Connectivity 6. Host and IOS Commands 7. Troubleshooting Methodologies 8. Troubleshooting Scenarios

Manajemen Jaringan

1 Data Communications and Network Management Overview

Subramanian, Mani (2000), *Network Management Principles and Practice*, Addison Wesley

- Telecommunications overview
- Data communications overview
- Evolution of converged networks
- Desktop processors and LAN technology
- Client-Server architecture in networking
- Internet and intranet
- Network communication protocols
- OSI and Internet s
- â€¢ Network components and technologies to be managed
- â€¢ Network Topologies: LAN and WAN
- â€¢ Wired LAN topology: Bus, Ring, Star, and Hybrid Hub
- â€¢ Wireless LAN
- â€¢ WAN topology: Mesh and Tree
- â€¢ Fixed and mobile wireless networks
- â€¢ Fiber networks
- â€¢ Ethernet LAN:
- â€¢ Physical media and MAC protocol
- â€¢ 10 and 100 Mbps; 1 and 10 Gbps Ethernet LAN
- â€¢ Switched and Duplex Ethernet LANs
- â€¢ Virtual LAN

2 Review of Information Network and Technology

Subramanian, Mani, "Network Management: Principles and Practice"

- â€¢ Token-ring LAN
- â€¢ FDDI
- â€¢ Network components
- â€¢ Bridges
- â€¢ Routers
- â€¢ Gateways
- â€¢ Circuit switching and packet switching
- â€¢ Transmission technology
- â€¢ Transmission media: Wired and Wireless
- â€¢ Transmission modes
- â€¢ Multiplexing: TDM and WDM
- â€¢ SONET and SDH
- â€¢ Multimedia networks and services
- â€¢ Standards, Models, and Language needed for network management
- â€¢ Network Models

3 Basic Foundations: Standards, Models, and Language

Subramanian, Mani, "Network Management: Principle and Practice".

- â€¢ OSI
- â€¢ Internet
- â€¢ TMN
- â€¢ IEEE 802
- â€¢ Web-based
- â€¢ Management communication protocols
- â€¢ SNMP
- â€¢ CMIP
- â€¢ XML
- â€¢ CORBA
- â€¢ ASN.1 language
- â€¢ Syntax
- â€¢ Macro
- â€¢ Basic encoding rule
- â€¢ Management application functions

Manajemen Jaringan

4	SNMPv1 Network Management: Organization and Information M	Subramanian, Mani, "Network Management: Principle and Practice".	- â€¢IETF SNMP standard - â€¢History - â€¢RFC, STD, and FYI - â€¢Organization Model - â€¢2- and 3-tier models - â€¢Manager and agent - â€¢Management messages - â€¢Structure of management information, SMI - â€¢Object type and instance - â€¢Scalar and aggregate managed objects - â€¢Management information base, MIB - â€¢NMS physical and virtual databases - â€¢IETF MIB-2 standard - â€¢Communication model: Administrative and messages - â€¢Administrative structure - â€¢Community-based model - â€¢Access policy - â€¢MIB view - â€¢Message PDU - â€¢SNMP protocol specifications - â€¢SNMP operations - â€¢SNMP MIB - â€¢SNMP functional model - â€¢Community-based security - â€¢SNMPv2 enhancements - â€¢Additional messages - â€¢Formalization of SMI - â€¢Get-bulk request and information-request - â€¢SNMP MIB modifications - â€¢Incompatibility with SNMPv1 - â€¢Proxy server - â€¢Bilingual manager - â€¢SNMPv3 features - â€¢Documentation architecture - â€¢Formalized SNMP architecture - â€¢Security - â€¢SNMP engine ID and name for network entity - â€¢SNMPv3 applications and primitives - â€¢SNMP architecture - â€¢Integrates the three SNMP versions - â€¢Message processing module - â€¢Dispatcher module - â€¢Future enhancement capability - â€¢User security model, USM - â€¢Derived from user ID and password - â€¢Authentication - â€¢Privacy - â€¢Message timeliness - â€¢View-based access control model, VACM - â€¢Configure set of MIB views for agent with contexts - â€¢Family of subtrees in MIB views - â€¢VACM process -Remote network monitoring, RMON -RMON1: Monitoring Ethernet LAN and token-ring LAN -RMON2: Monitoring upper protocol layers -Generates and sends statistics close to subnetworks to central NMS -RMON MIBs for RMON group objects
5	SNMPv1 Network Management: Communication and Functional M	Subramanian, Mani, "Network Management: Principle and Practice".	
6	SNMP Management: SNMPv2	Subramanian, Mani, "Network Management: Principle and Practice".	
7	SNMP Management: SNMPv3	Subramanian, Mani, "Network Management: Principle and Practice".	
8	SNMP Management: RMON	Subramanian, Mani, "Network Management: Principle and Practice".	

Manajemen Jaringan	9	Telecommunications Management Network	- Telecommunications Management Network, TMN - Concept of Operations Support System, OSS - TMN conceptual model includes: - Customers - Service providers - Network - Operations support systems, OSSs - System operators - TMN standards and documentation - TMN architecture - Functional - Physical - Informational - TMN service management architecture - Network element - Element management - Network management - Service management - Business management - TMN service management - Operations, Administration, Maintenance, Provisioning; OAMP - TMN implementation methodologies - COMNIPoint
	10	Network Management Tools, Systems, and Engineering	- Subramanian, Mani, "Network Management: Principle and Practice". - Status monitoring tools - Traffic monitoring tools - Route monitoring tools - Functional Requirements - Scalability - Heterogeneity - Geographic spread - Bursty Load - Real-time response - Batch processing - Diverse users - Local and remote management - Ease of use - Security - Data management
	11	Network Management System Design	Subramanian, Mani, "Network Management: Principle and Practice".

			• Network management and system management • Network management • Configuration • Fault • Performance • Security • Accounting • Configuration management • Configuration management • Service/Network provisioning • Inventory management • Fault management • Fault detection and isolation • Correlation techniques for root cause analysis • Performance management • Performance metrics • Data monitoring • Problem isolation • Performance statistics • Security management • Security policies and procedures • Security threats • Firewall • Cryptography: keys, algorithms, authentication, and authorization schemes • Secure message transfer methods • Accounting management • Broadband WAN segment • IP (has been dealt with earlier) • ATM • MPLS • Optical and MAN feeder network • ATM • Virtual path/virtual circuit (VP-VC) operation • Real-time and non-real-time function for broadband service • ATM as public and private switched networks • Emulated LAN configuration • ATM management: M1, M2, M3, and M4 interfaces • ATM digital exchange interface management • CMPLS • Possesses rich features of IP and good performance of ATM • Basic principles of label switching • Label switched path, LSP • Traffic engineering • MPLS OAM • Service level management • MPLS MIBs • MPLS TE MIBs • MPLS example • Optical and MAN feeder network • SONET-based MAN • SONET transport hierarchy • SDH and (D)WDM network • SDH management • WDM management • Introduction to discrete math. • Syllabus explanation. • Sets. • Venn Diagram.
Manajemen Jaringan	12	Network Management Applications	Subramanian, Mani, "Network Management: Principles and Practice".
	13	Broadband Network Management: WAN	Subramanian, Mani, "Network Management: Principle and Practice"
	14	Broadband Network Management: WAN Part II	
Matematika Diskrit	1	Introduction to Discrete Math.	Subramanian, Mani, "Network Management: Principle and Practice". Textbook, Ch. 1 (Sets)

Matematika Diskrit	2	Set (Cont'd) and logics (intro)	Textbook Ch. 1	Sets (disc. on assignment 1, venn diagram etc). Logics (intro). Preposition, Conditional, Contrapositive, Converse. Arguments. Quantifiers. Universal Quantifiers. Existensial Quantifiers. Direct proof. Indirect proof. Proof By Contradiction. Mathematical induction. Steps of mathematical induction. Strong form of induction. - Functions. - Injective vs Surjective vs Bijective - Sequence. - Strings. - Relations. - Infinite sequences. - Cardinality / countability of infinite sequences. - Recurring Relations. - Homogeneous equation and characteristic equation. - Solution to recurring sequences. - Mid Exam Review. - Introduction to Number Theory. GCD and LCM Euclidean Algorithm and Extended Euclidean Algorithm 2 Number Systems Decimal, Binary, and Hexadecimal Conversion Techniques 3 One's and Two's Complement Representation and Arithmetic Operations 4 Modular Arithmetic Fast Modular Exponentiation Applications in Cryptography 5 Prime Numbers and Factorization 6 Applications and Examples Euclidean algorithm (extended). Fast modular exponentiation. FME. Chinese Remainder Theorem. Cryptography. Combinatoric and counting using Pascal's triangle. Additive and multiplicative rules. Fundamental counting principles. Permutation: linear, circular. Combinations. Pigeonhole principles.
	3	Arguments, Quantifiers, Proof.	Textbook Ch 1.5, Ch. 2	
	4	Proof.	Textbook Ch. 2	
	5	Mathematical induction.	Textbook Ch. 2.4 - 2.5	
	6	Function, Sequence, Strings	Textbook Ch. 3	
	7	Sequences.	Textbook	
	8	Review Mid-Exam, Intro to number theory.	Textbook.	
	9	Number Theory		
	10	Number Theory	Slide, textbook, notes	
	11	Number theory (Continued)	Slide, notes	
	12	Counting.	Textbook, slides, notes.	
	13	Counting Theory	Slide, Notes, Textbook.	
	14	Graphs	Slides, notes, textbook.	
			Slides, notes.	
Pemrograman Berorientasi Objek	1	Introduction	Gaddis	Introduction - OOP concept - Inheritance, Polymorphism, encapsulation, abstraction 1. Pendahuluan 2. Definisi PBO 3. PBO vs Prosedural 4. Keunggulan PBO 5. Empat Pilar Utama dalam PBO 6. Tanya & Jawab
		Week#1 - Pengenalan Pemrograman Berorientasi Objek	-	

Pemrograman Berorientasi Objek	2	Java Fundamentals	Deitel	Java Fundamentals - primitive and class data type. - rank of data type. - static method 1. Class 2. Object 3. Contoh Class dan Object 4. Attribute 5. Method 6. Implementasi Class dan Object Berdasarkan Contoh 7. Latihan Java classes and Objects Field and method Passing argument Package and import 1. Resume Materi Kelas dan Objek 2. Pendahuluan 3. Java Scanner 4. Implementasi Scanner pada Java 5. Input Tokens 6. Latihan Desicion structure If then...else Elseif.. Switch Ternary statement
		Week#2 - Class dan Object	Iskandar, Ade Rahmat, 2020, Menguasai Pemrograman Berorientasi Objek, Jakarta, Informatika.	
	3	Java classes and Objects	Deitel	1. Resume Scanner 2. Variables 3. Jenis Variabel 4. Penamaan Variabel pada Java 5. Method 6. Deklarasi Method 7. Penulisan Method dan Pemanggilan Method 8. Jenis Method 9. Latihan
		Week#3 - Java Scanner	Munir, Rinaldi. 2024. Pengantar Pemrograman dengan Bahasa Java. Informatika STEI ITB. CSE 142 Lecture Notes: Interactive Programs with Scanner. University of Washington http://www.w3resource.com/java-tutorial/java-class-methods-instance-variables.php http://javajee.com/variables-and-data-types-in-java Brucker, James. Variables other sources from Google	
	4	Desicion structure	Deitel	Loop and File -Read/write file -for loop -while loop -do..while loop
		Week#4 - Variables dan Method	Munir, Rinaldi. 2024. Pengantar Pemrograman dengan Bahasa Java. Informatika STEI ITB. CSE 142 Lecture Notes: Interactive Programs with Scanner. University of Washington http://www.w3resource.com/java-tutorial/java-class-methods-instance-variables.php http://javajee.com/variables-and-data-types-in-java Brucker, James. Variables other sources from Google	1. Resume Variable dan Methods 2. Definisi Access Modifier 3. Jenis Access Modifier 4. Access Modifier Privelege 5. Contoh Kasus 6. Latihan
	5	Loop and File	Deitel	
		Week#5 - Access Modifier	Munir, Rinaldi. 2024. Pengantar Pemrograman dengan Bahasa Java. Informatika STEI ITB. CSE 142 Lecture Notes: Interactive Programs with Scanner. University of Washington http://www.w3resource.com/java-tutorial/java-class-methods-instance-variables.php http://javajee.com/variables-and-data-types-in-java Brucker, James. Variables other sources from Google	
	6	Second look class	Deitel	Second look class

Pemrograman Berorientasi Objek

6	Week#6 - Constructors	Munir, Rinaldi. 2024. Pengantar Pemrograman dengan Bahasa Java. Informatika STEI ITB. CSE 142 Lecture Notes: Interactive Programs with Scanner. University of Washington http://www.w3resource.com/java-tutorial/java-class-methods-instance-variables.php http://javajee.com/variables-and-data-types-in-java Brucker, James. Variables Jadoon, Rab. Introduction to Java Constructors. Comsat University. Islamabad. Pakistan other sources from Google	1. Review Access Modifier 2. Pengantar Constructor 3. Contoh Constructor pada Java 4. Jenis Constructor 5. Implementasi Default Constructor 6. Implementasi Non-Arg Constructor 7. Implementasi Parameterized Constructor
7	Array and Arraylist	Deitel	Array and Arraylist 1. Review Kelas dan Objek 2. Review Method dan Variabel 3. Review Access Modifier 4. Review Constructor
	Week#7 - Review untuk UTS	-	Text processing and wrapper class
8	Text Processing	Gaddis	1. Pengantar Encapsulation 2. Penerapan Encapsulation 3. Set Method 4. Get Method 5. Latihan 6. Tugas 7. Inheritance 8. Base class & child class. 9. Overriding method.
	Week#9 - Encapsulation	1. Iskandar, Ade Rahmat, 2020, Menguasai Pemrograman Berorientasi Objek, Jakarta, Informatika. 2. Head First Object Oriented Design and Analysis, 1st edition, 2006, Brett D. McLaughlin, Gary Pollice, David West, O'Reilly Media 3. JavaTM How to Program, 9th, 2012, Prentice Hall 4. Head First Java, 2nd edition, 2008, Bert Bates and Kathy Sierra, O'Reilly	
9	Inheritance	Deitel	
	Week#8 - Inheritance	Munir, Rinaldi. 2024. Pengantar Pemrograman dengan Bahasa Java. Informatika STEI ITB. CSE 142 Lecture Notes: Interactive Programs with Scanner. University of Washington http://www.w3resource.com/java-tutorial/java-class-methods-instance-variables.php http://javajee.com/inheritance-in-java Brucker, James. Variables Jadoon, Rab. Introduction to Java Constructors. Comsat University. Islamabad. Pakistan other sources from Google	1. Pengantar Inheritance 2. Tipe Inheritance 3. Single Inheritance 4. Multi-Level Inheritance 5. Hierarchical Inheritance 6. Latihan 7. Penugasan
10	Guest Lecture	Pemrograman Berorientasi Objek - Modul Praktikum	Guest Lecture
	Week#10 - Polymorphism	1. Iskandar, Ade Rahmat, 2020, Menguasai Pemrograman Berorientasi Objek, Jakarta, Informatika. 2. Head First Object Oriented Design and Analysis, 1st edition, 2006, Brett D. McLaughlin, Gary Pollice, David West, O'Reilly Media 3. JavaTM How to Program, 9th, 2012, Prentice Hall 4. Head First Java, 2nd edition, 2008, Bert Bates and Kathy Sierra, O'Reilly	1. Pengantar Polymorphism 2. Method Overloading 3. Method Overriding 4. Keyword "Super" 5. Latihan pada Java
11	Exception handling		Exception handling. FileNotFoundException. ArrayIndexOutOfBoundsException.
	Week#12 - Interface	Gaddis	1. Pengantar Interface 2. Perbedaan Utama Interface dan Class Umum 3. Ciri-Ciri Interface 4. Deklarasi Interface 5. Implementasi Interface 6. Latihan 7. Tugas
		1. Jadoon, Rab. Introduction to Java Constructors. Comsat University. Islamabad. Pakistan 2. other sources from Google 3. Pemrograman Berorientasi Objek - Modul Praktikum	

Pemrograman Berorientasi Objek	12	GUI 1	Tonny Gaddis	GUI 1 Dialog box LayOut Manager (Flow, Border and Grid layout) JFrame and JPanel
		Week#11 - Abstract Class		1. Pengantar Abstract Class 2. Struktur Dasar Abstract Class 3. Object Abstract Class 4. Method Abstract 5. Latihan 6. Tugas GUI-2 Tiap kelompok presentasi dan live demo aplikasi karya Quiz Melakukan revisi pada Project Akhir UAS
	13	GUI-2 Week#13 - Presentasi Project UAS	Pemrograman Berorientasi Objek - Modul Praktikum - UMS Gaddis	
	14	Quiz Week#14 - Revisi Project	- Gaddis -	
Pemrograman Visual	1	Introduction		Introduction - OOP - Visual basic programming language - event driven / listener Dive into visual studio 2022 Introduction to Visual Basic programming 1 Introduction to visual programming 2 Control statement 1 if..then it..then...else elseif nested if.
	2	Dive into visual studio 2022	Dietel	
	3	Introduction to Visual Basic programming 1	Gaddis	
	4	Introduction to visual programming 2	Gaddis	
	5	Control statement 1		
	6	Control statement 2	Gaddis	
	7	Control statement 3	Gaddis	
	8	Method 1		
	9	Method 2	Gaddis	
	10	Array 1	Deitel	
	11	Array 2	Deitel	
	12	Object Oriented Programming	Gaddis	
Proses Sinyal Dijital	13	Mock up project semester presentation	Deitel	
	14	Presentasi	Gaddis	
	1	Introduction to DSP	Deitel	
		Week#1 - Pengenalan Pengolahan Sinyal Digital	Textbook, Slides. Jonathan Stein, & Digital Signal Processing: A Computer Science Perspective& John Wiley & Sons, 2000 Unofficial url: dsp-book.narod.ru/DSPCSP/toc.htm Unofficial pdf (55 MB pdf book): https://files.bakrie.ac.id/s/XGnTjLxLcM78QTT Hardcopy is also available in our Library Steven Smith, & The Scientist and Engineer's Guide to Digital Signal Processing& California Technical Publishing. Downloadable at http://www.dspguide.com/pdfbook.htm	1. Pengantar 2. Sinyal 3. Jenis Sinyal 4. Pengolahan Sinyal 5. Aplikasi Pengolahan Sinyal

Proses Sinyal Dijital

2	Signals representation and manipulation.	Textbook, Slides. [1] Steven W. Smith. The Scientist and Engineer's Guide to Digital Signal Processing. California Technical Publishing, 1997. URL http://www.dspguide.com/pdfbook.htm. [2] Sanjit K. Mitra and James F. Kaiser, editors. Handbook for Digital Signal Processing. John Wiley & Sons, 1993.	Signals representation. Basic signals. Aliasing. Signals manipulations. 1. Apa itu Sinyal? 2. Karakteristik Sinyal 3. Klasifikasi Sinyal 4. Latihan Pembuatan Sinyal
3	Signal manipulation (cont'd) and systems	Textbook, Slides. Alan V. Oppenheim and Ronald W. Schafer. Discrete-Time Signal Processing. Prentice-Hall, 2nd edition, 1999. John G. Proakis and Dimitris G. Manolakis. Digital Signal Processing: Principles, Algorithms, and Applications. Macmillan, second edition, 1992. Paul A. Lynn and Wolfgang Fuerst. Introductory Digital Signal Processing with Computer Applications. John Wiley & Sons, revised edition, 1994. Budi Ardiperdana. Pengolahan Sinyal Digital.	Calculations of signal manipulations. Systems. 1. Sistem 2. Klasifikasi Sistem 3. DSP vs ASP 4. Sampling 5. Kuantisasi 6. Latihan Convolution sum basic. Graphical representation of convolution. Array algorithm.
4	Convolution.	Textbook, slide, notes. 1. Alan V. Oppenheim and Ronald W. Schafer. Discrete-Time Signal Processing. Prentice-Hall, 2nd edition, 1999. 2. John G. Proakis and Dimitris G. Manolakis. Digital Signal Processing: Principles, Algorithms, and Applications. Macmillan, second edition, 1992. 3. Paul A. Lynn and Wolfgang Fuerst. Introductory Digital Signal Processing with Computer Applications. John Wiley & Sons, revised edition, 1994. 4. Steven W. Smith. The Scientist and Engineer's Guide to Digital Signal Processing. California Technical Publishing, 1997. URL http://www.dspguide.com/pdfbook.htm. 5. Sanjit K. Mitra and James F. Kaiser, editors. Handbook for Digital Signal Processing. John Wiley & Sons, 1993.	1. Resume Sampling 2. Latihan Sampling Pembahasan Soal 3. Domain Waktu 4. Latihan Soal Domain Waktu 5. Domain Frekuensi 6. Simpulan
5	Impulse Response	Textbook, slides, notes. 1. Alan V. Oppenheim and Ronald W. Schafer. Discrete-Time Signal Processing. Prentice-Hall, 2nd edition, 1999. 2. John G. Proakis and Dimitris G. Manolakis. Digital Signal Processing: Principles, Algorithms, and Applications. Macmillan, second edition, 1992. 3. Paul A. Lynn and Wolfgang Fuerst. Introductory Digital Signal Processing with Computer Applications. John Wiley & Sons, revised edition, 1994. 4. Steven W. Smith. The Scientist and Engineer's Guide to Digital Signal Processing. California Technical Publishing, 1997. URL http://www.dspguide.com/pdfbook.htm. 5. Sanjit K. Mitra and James F. Kaiser, editors. Handbook for Digital Signal Processing. John Wiley & Sons, 1993.	Impulse response. Calculation for FIR system (direct calculation). Calculation for IIR system (unforced input with initial condition).
	Transformasi Fourier		1. Review Domain Waktu dan Domain Frekuensi 2. Sinyal Diskrit 3. Pengantar Transformasi Fourier 4. Jenis Transformasi Fourier 5. Discrete Fourier Transform (DFT) 6. Latihan Menghitung Spektrum Frekuensi dengan DFT 7. Simpulan
6	Impulse respons (cont'd)	Textbook, slides, notes	IR calculation for ARMA systems.
	Review Materi	-	Linearity principles in calculating IR. Melakukan diskusi dan review materi Pra Mid Term

Proses Sinyal Dijital	7	Fast Fourier Transform (FFT)	1. Alan V. Oppenheim and Ronald W. Schaffer. Discrete-Time Signal Processing. Prentice-Hall, 2nd edition, 1999. 2. John G. Proakis and Dimitris G. Manolakis. Digital Signal Processing: Principles, Algorithms, and Applications. Macmillan, second edition, 1992. 3. Paul A. Lynn and Wolfgang Fuerst. Introductory Digital Signal Processing with Computer Applications. John Wiley & Sons, revised edition, 1994. 4. Steven W. Smith. The Scientist and Engineer's Guide to Digital Signal Processing. California Technical Publishing, 1997. URL http://www.dspguide.com/pdfbook.htm. 5. Sanjit K. Mitra and James F. Kaiser, editors. Handbook for Digital Signal Processing. John Wiley & Sons, 1993.	1. Review Materi DFT 2. Pengantar Fast Fourier Transform 3. Rumus FFT 4. Diagram Butterfly 5. Gambar Diagram Butterfly untuk $N = 4$ dan $N = 8$ 6. Contoh Perhitungan FFT 7. Latihan Soal
	8	LTI system responses (Quiz) Review Midterm Exam	Textbook, slides, notes. Textbook, clides	- LTI system output. - LTI system impulse responses. Review Midterm Exam 1. Review FFT 2. Pengantar Spektrum Frekuensi 3. Spektrum Amplitudo dan Spektrum Daya 4. Panjang Sinyal 5. Zero Padding 6. Latihan Fourier analysis types. Property of Fourier Analysis. 1. Pengantar Spektrogram 2. Sinyal Stasioner dan Non-Stasioner 3. Penggambaran Spektrogram 4. Perbedaan Spektrum Biasa dan Spektrogram 5. Latihan
	9	Week#8 - Spektrum Frekuensi Fourier Analysis.	Pah, Nemuel Daniel. 2018. Pemrosesan Sinyal Digital. Graha Ilmu. Yogyakarta Slide, colab	1. Pengantar Spektrum 2. Sinyal Stasioner dan Non-Stasioner 3. Penggambaran Spektrogram 4. Perbedaan Spektrum Biasa dan Spektrogram 5. Latihan
	10	Week#9 - Spektrogram Inverse Z-Transform	Pah, Nemuel Daniel. 2018. Pemrosesan Sinyal Digital. Graha Ilmu. Slide, notes.	Inverse Z-Transform Filtering. System simulation using J-DSP. 1. Pengantar Sinyal Suara 2. Voice dan Unvoiced 3. Frekuensi pada Voiced Speech 4. Identifikasi Pitch f_0 Exercise on: CTFS. CTFT. DTFS. Latihan teori dan praktikum coding pada m Sinyal Suara
	11	Week#10 - Analisis Sinyal Suara Fourier Representation (Exercise)	Pah, Nemuel Daniel. 2018. Pemrosesan Sinyal Digital. Graha Ilmu. Slides, notes, textbook.	1. Review Analisis Sinyal Suara 2. Pengantar Sistem Diskrit 3. Sistem Diskrit bentuk Impulse Response 4. Latihan Soal Forward Z-Transform. Region of Convergence. Inverse Z-Transform. Direct computation using infinite summation. Look up table. 1. Pengantar 2. Sistem Diskrit dengan Bentuk Transfer Function 3. Latihan Review materials for final exam. Transfer function. Inverse Z-Transform. Latihan mengerjakan soal sistem diskrit
	12	Week#11 - Latihan soal Spektrogram dan Analisis Sinyal Suara Week#12 - Sistem Diskrit	Pah, Nemuel Daniel. 2018. Pemrosesan Sinyal Digital. Graha Ilmu. Slide, notes.	1. Review Analisis Sinyal Suara 2. Pengantar Sistem Diskrit 3. Sistem Diskrit bentuk Impulse Response 4. Latihan Soal Forward Z-Transform. Region of Convergence. Inverse Z-Transform. Direct computation using infinite summation. Look up table. 1. Pengantar 2. Sistem Diskrit dengan Bentuk Transfer Function 3. Latihan Review materials for final exam. Transfer function. Inverse Z-Transform. Latihan mengerjakan soal sistem diskrit
	13	Z-Transform Inverse Z-Transform	Slide, notes, exercise. Pah, Nemuel Daniel. 2018. Pemrosesan Sinyal Digital. Graha Ilmu.	1. Pengantar 2. Sistem Diskrit dengan Bentuk Transfer Function 3. Latihan Review materials for final exam. Transfer function. Inverse Z-Transform. Latihan mengerjakan soal sistem diskrit
	14	Week#13 - Transfer Function Final Destination: UAS	Slide, videos, notes.	Students are able to explain the main concept of software engineering
	1	Week#14 - Latihan Sistem Diskrit Bentuk Transfer Function Introduction to software engineering	Sommerville, Ian. (2016). Software Engineering (Ed. 10th). Boston: Pearson Menjelaskan Sistem Perkuliahan dan Menjelaskan Pengertian dan contoh Rekayasa Perangkat Lunak (RPL)/ Software Engineering	Mahasiswa diajarkan tentang penertian dan Contoh produk Rekayasa Perangkat Lunak
		Rekayasa perangkat Lunak		

Rekayasa perangkat Lunak	2	Rekayasa Perangkat Lunak	Menjelaskan tentang software dan jenis2 software	Mahasiswa dijelaskan tentang software dan jenis2 software dan cara pengembangan software
		Requirement Engineering	Pressman, R. (2010). <i>Software Engineering: Practitioner's Approach, 7Edition.</i> McGraw-Hill.	Students are able to explain the main concept of requirement engineering.
	3	Rekayasa Perangkat Lunak	Sommerville, I. (2011). <i>Software Engineering, 9th edition, Addison-Wesley.</i>	Students are able to explain the functionality of Software Requirement Specification.
		Software Design	Mahasiswa diajarkan bagaimana merancang perangkat lunak dengan menggunakan model perancangan perangkat lunak	Mahasiswa diajarkan merancang perangkat lunak dengan model Waterfall
	4	Rekayasa Perangkat Lunak	Sommerville I, <i>Software Engineering</i>	Students are able to translate the software requirement into software design
		Software Design	Mahasiswa diajarkan penggunaan Tool untuk merancang Software	Mahasiswa diajarkan Menggunakan Tools UML dalam merancang Software
	5	Coding phase	Pressman R. <i>Software Engineering</i>	Students are able to create software design description (SDD) and calculate the main effort based on
		Rekayasa Perangkat Lunak	Sommerville, <i>Software Engineering</i>	Students are able to translate design into code.
	6	Menjelaskan cara merancang Perangkat Lunak dengan UML	Menjelaskan Tools perancangan Perangkat Lunak UML, sub bagian Use Case	Mahasiswa diajarkan untuk merancang RPL dengan Menggunakan UML yaitu : Use Case
		Software Construction and Testing	Mahasiswa diajarkan bagaimana cara merancang software dengan UML	Mahasiswa diajarkan cara merancang Software dengan UML; yaitu merancang software Aplikasi
		Software Maintenance and Coni		1) Students are able to explain the main role of software construction (code)
	7	Rekayasa Perangkat Lunak; Activity Diagram	Sommerville, I. <i>Software Engineering, 9th edition, Addison-Wesley.</i>	2) Students are able to explain the objective of software testing in SE.
		Software Quality	Menjelaskan bagian dari merancang RPL dengan UML; Activity Diagram	3) Students are able to explain the main role of software maintenance.
	8	Rekayasa Perangkat Lunak	Sommerville, software engineering	4) Students are able to explain the main role of software configuration.
		Workshop on software engineering: Requirement	Merancang Perangkat Lunak	Mahasiswa diajarkan merancang RPL dengan UML; Activity Diagram
Sistem Basis Data Lanjut	9	Rekayasa Perangkat Lunak	Sommerville, I, <i>Software Engineering</i>	Students are able to explain the testing procedure and able to create test plan.
		Workshop on Software Requirements	Perancangan Sistem dengan menggunakan UML	Mahasiswa diajarkan merancang Perangkat Lunak dengan metode berorientasi Object, yaitu dengan
	10	Guest Lecture	Roger Pressman, <i>Software Engineering</i>	Students are able to explain the main methods in gathering software requirements
		Perancangan Sistem dengan menggunakan DFD (Data Flow Diagram	-	Mahasiswa diajarkan bagaimana cara untuk merancang Sistem dengan menggunakan UML
	11	Presentasi SRS	Menjelaskan Bagaimana cara merancang Sistem Informasi dengan DFD (Data Flow Diagram)	Students are able to write a formal software requirements
		Rekayasa Perangkat Lunak	Software engineering	Mahasiswa menghadiri guest lecture.
	12	Merancang Sistem Rekayasa Perangkat Lunak dengan DFD	Perancangan Sistem dengan menggunakan DFD	Mahasiswa Diajarkan bagaimana merancang Sistem Informasi dengan menggunakan DFD (Data Flow Diagram)
		Workshop on software design	Merancang Sistem Rekayasa Perangkat Lunak dengan DFD	Students are able to present working progress on documenting software requirement
	13	Rekayasa Perangkat Lunak	Ian Sommerville, <i>software engineering</i>	Mahasiswa diajarkan bagaimana merancang Sistem dengan menggunakan Data Flow Diagram
		Workshop on software coding	Merancang System dengan DFD	Mahasiswa diajarkan Bagaimana cara untuk merancang sistem dengan DFD
	14	Final Presentation	Sommerville, <i>software engineering</i>	Students are able to present the main progress of developing the software
		Rekayasa Perangkat Lunak	Sommerville I, <i>software engineering</i>	Mahasiswa diajarkan bagaimana cara merancang sistem dengan DFD
			Riview	Students are able to present the working progress of software development: coding phase
			1Thomas Connolly, Carolyn Begg; <i>Database Systems: A Practical Approach to Design, Implementation, and Management; 6th eds; International Edition; Pearson USA (2015)</i>	students are able to present the final software and its documentation
Sistem Terdistribusi	1	Course Introduction & Introduction to object-oriented database:3	2Hoffer, J. A., Ramesh, V., & Topi, H. (2016). <i>Modern database management 12 eds. Chapter 14</i>	Dilakukan Riview Materi Perkuliahan 9 s/d 15 untuk menyegarkan ingatan Mahasiswa menghadapi
		Introduction to advanced database	connolly, <i>database management system</i>	a. Building Learning Commitment
	2	Object Oriented data modeling	Hoffer, <i>Database Management System</i>	b. An Overview of Data Model
	3	Object Oriented data modeling	Hoffer J, <i>Database Management System</i>	c. An overview of Object-Oriented Database System
	4	Object Oriented data modeling	Hoffee J, <i>Database Management System ch14</i>	
	5	Object-Query Language (OQL)	the definitive guide to db4o	students are able to explain the others type of database system (others than RDBMS)
	6	ODL, OQL, transaction & Java Apps	Horning, H., Horning, R., Eldich, S., & Paterson, J. (2006). <i>The Definitive Guide to db4o. Apress.</i>	Students are able to create a data model based on OODB paradigm
	7	Review on object oriented database	hoffer, <i>database management system</i>	Students are able to create an Object Oriented data modeling based on specific cases.
	8	XML	hoffee, <i>modern database system</i>	Students are able to create an object-oriented data model and using db4o
	9	xml, xpath, xquery and web services		Students are able to create ODL and OQL (Native Query, QBE as well as SODA query)
	10	Graph Database	Pro RESTful APIs: Design, Build and Integrate with REST, JSON, XML and JAX-RS	Students are able to create an Object Query Language (OQL): basic SODA Query on db4o (Java-based);
	11	NoSQL	Jeff Hoffer, Ramesh V and Heikki Topi. 2016. <i>Modern Database Management, 12th Ed, Pearson.</i>	Students are able to create an Object Query Language (OQL): advanced SODA Query on db4o (Java-)
	12	NoSQL - Document DB	MongoDB	Students are able to design and create an object oriented database using db4o and Java
	13	GraphDB use case	MongoDB	Students are able to explain the main concept of XML and its application
	14	Presentasi proyek akhir NoSQL	introduction to graph database	Students are able to create cml and make query using xpath, xquery and web services: restful
	1	Introduction to distributed system	Connolly, <i>Database management system</i>	Students are able to explain the main concept of graph database
			Tannenbaum	Students are able to explain the main concept of document database
				Students are able to create queries to retrieve data
				Students are able to enrich the main conceptual of graphDB with related use cases
				Students are able to present the latest progress on final project
				Introduction to distributed system

Sistem Terdistribusi

1 Introduction to Distributed Systems

â€¢ Tanenbaum. Andrew S, Maarten Van Steen, â€¢ Distributed Systems: Principles and Paradigms, 3rd Ed.â€¢ Pearson Education, Inc., 2017 (Chapter 1).
 â€¢ Coulouris, Dollimore, Kindberg, â€¢ Distributed Systems: Concepts and Design 5th editionâ€¢ Addison Wesley: 2005 (Chapter 1).
 â€¢ Velte. Toby, Anthony Velte, Robert Elsenpeter, â€¢ Cloud Computing, A Practical Approachâ€¢ McGraw Hill Osborne Media., 2012 (Chapter 1, 4).
 â€¢ Vitillo . Roberto, â€¢ Understanding Distributed Systems: What every developer should know about large distributed applications, 2nd Ed.â€¢ 2022 (Chapter 1)

1. Definition of Distributed System
2. Goals of Distributed Systems
3. Types of Distributed Systems
4. Examples of Distributed Systems
5. Trends in Distributed Systems
6. Focus on Resource Sharing
7. Challenges in Distributed Systems
8. Case Study: The World Wide Web
9. Anatomy of a Distributed Systems
10. Cloud Computing Basics
11. Applications
12. Intranets and the Cloud
13. First movers in the Cloud
14. The Business Case for Going to the Cloud
15. Cloud Computing Services
16. How Those Applications Help Your Business
17. Deleting Your Datacenter
18. Chapter Summary

2 Architectures

1. Tanenbaum. Andrew S, Maarten Van Steen, â€¢ Distributed Systems: Principles and Paradigms, 3rd Ed.â€¢ Pearson Education, Inc., 2017.
2. Coulouris, Dollimore, Kindberg, â€¢ Distributed Systems: Concepts and Design 5th editionâ€¢ Addison Wesley: 2005.
3. Velte. Toby, Anthony Velte, Robert Elsenpeter, â€¢ Cloud Computing, A Practical Approachâ€¢ Mc-Graw Hill Osborne Media., 2012.
4. Vitillo. Roberto, â€¢ Understanding Distributed Systems: What every developer should know about large distributed applications, 2nd Ed.â€¢ 2022.
5. Hunter II, Thomas, â€¢ Distributed Systems with Node.js: Building Enterprise-Ready Backend Servicesâ€¢ Oâ€™Reilly Media Inc., 2021.
6. Red Hat® Team - RHEL RH134 - EN, â€¢ Red Hat System Administration IIâ€¢ 2024, <https://www.redhat.com/en/services/training/red-hat-academy> : 2024.
7. Red Hat® Team,â€¢ DO101-Introduction to OpenShift Applicationsâ€¢ Red Hat Academy Course, <https://rha.ole.redhat.com/rha/app/summary> : 2024.

- (a) Architectural styles
- (b) Middleware Organizations
- (c) System architectures
- (d) Example Architectures
- (e) Physical Models
- (f) Architectural Models
- (g) Fundamental Models
- (h) Architectures versus middleware
- (i) Self-management in distributed systems
- (j) Hardware and Infrastructure of Cloud Computing
- (k) Clients in Cloud Computing
- (l) Security
- (m) Network
- (n) Services
- (o) Chapter Summary

architecture style

Tannenbaum

architecture style
 Presentasi tugas grup 1 dan 2.

3 Processes

1. Tanenbaum. Andrew S, Maarten Van Steen, â€¢ Distributed Systems: Principles and Paradigms, 3rd Ed.â€¢ Pearson Education, Inc., 2017 (Chapter 3).
2. Coulouris, Dollimore, Kindberg, â€¢ Distributed Systems: Concepts and Design 5th editionâ€¢ Addison Wesley: 2005 (Chapter 7.4).

Tannenbaum

1. Threads
2. Virtualization
3. Clients
4. Servers
5. Code Migration
6. Process and Threads
7. Chapter Summary

4 Communication

Tannenbaum

Processes
 Communication
 Basic OSI model.
 Transient vs Persistent
 asynchronous vs synchronous

Sistem Terdistribusi	4	Communication, Inter process and indirect communication	1. Tanenbaum. Andrew S, Maarten Van Steen, <i>Distributed Systems: Principles and Paradigms</i>, 3rd Ed. <i>Pearson Education, Inc., 2017 (Chapter 4)</i>. 2. Coulouris, Dollimore, Indberg, <i>Distributed Systems: Concepts and Design 5th edition</i> <i>Addison Wesley: 2005 (Chapter 4, 6)</i>. 3. Velte. Toby, Anthony Velte, Robert Elsenpeter, <i>Cloud Computing, A Practical Approach</i> <i>McGraw Hill Osborne Media., 2012 (Chapter 8)</i>.	- Foundations - Remote Procedure Call - Message-oriented Communication - Multicast Communication - Standards and Applications - Client - Infrastructure - Service - Introduction to Interprocess Communication - The API for the Internet Protocols - External data representation and marshalling - Network Virtualization: Overlay Networks - Introduction to Indirect Communication - Group Communication - Publish-Subscribe Systems - Message Queues - Shared Memory Approaches - Chapter Summary - Names, Identifiers and Addresses - Flat Naming - Structured Naming - Attribute-based Naming - Introduction to Name Services - Names Services and the Domain Name System - Directory Services - Case Study: The Global Name Service - Chapter Summary
	5	Naming and Name Services	1. Tanenbaum. Andrew S, Maarten Van Steen, <i>Distributed Systems: Principles and Paradigms</i>, 3rd Ed. <i>Pearson Education, Inc., 2017 (Chapter 5)</i>. 2. Coulouris, Dollimore, Kindberg, <i>Distributed Systems: Concepts and Design 5th edition</i> <i>Addison Wesley: 2005 (Chapter 13)</i>.	- Names, Identifiers and Addresses - Flat Naming - Structured Naming - Attribute-based Naming - Introduction to Name Services - Names Services and the Domain Name System - Directory Services - Case Study: The Global Name Service - Chapter Summary
		Presentasi tugas 2	Tannenbaum	Presentasi Communication process - Clock Synchronization - Logical Clocks - Physical Clocks - Vector Clocks - Mutual Exclusion - Election Algorithms - Leader Election - Transactions and Concurrency Control - Location Systems - Distributed Event Matching - Gossip-based Coordination - Coordination and Agreement in Group Communication - Consensus and Related Problems - Cloud Storage - Chapter Summary
	6	Coordination	1. Tanenbaum. Andrew S, Maarten Van Steen, <i>Distributed Systems: Principles and Paradigms</i>, 3rd Ed. <i>Pearson Education, Inc., 2017 (Chapter 6)</i>. 2. Coulouris, Dollimore, Kindberg, <i>Distributed Systems: Concepts and Design 5th edition</i> <i>Addison Wesley: 2005 (Chapter 15.4)</i>. 3. Velte. Toby, Anthony Velte, Robert Elsenpeter, <i>Cloud Computing, A Practical Approach</i> <i>McGraw Hill Osborne Media., 2012 (Chapter 7, 8.1, 8.2, 9)</i>. 4. Vitillo. Roberto, <i>Understanding Distributed Systems: What every developer should know about large distributed applications</i>, 2nd Ed. <i>2022 (Chapter 16)</i>	- Leader Election - Transactions and Concurrency Control - Location Systems - Distributed Event Matching - Gossip-based Coordination - Coordination and Agreement in Group Communication - Consensus and Related Problems - Cloud Storage - Chapter Summary
		Naming	Tannenbaum	Naming - flat naming - structured naming - attribute-based naming
	7	Consistency and Replication	1. Tanenbaum. Andrew S, Maarten Van Steen, <i>Distributed Systems: Principles and Paradigms</i>, 3rd Ed. <i>Pearson Education, Inc., 2017 (Chapter 7)</i>. 2. Coulouris, Dollimore, Kindberg, <i>Distributed Systems: Concepts and Design 5th edition</i> <i>Addison Wesley: 2005 (Chapter 18.5)</i>. 3. Vitillo. Roberto, <i>Understanding Distributed Systems: What every developer should know about large distributed applications</i>, 2nd Ed. <i>2022 (Chapter 10.1, 10.3)</i>	- Introduction of Consistency and Replication - Data-centric Consistency Models - Client-centric Consistency Models - Replica Management - State Machine Replication - Consistency Protocols - Consistency Models - Transactions with Replicated Data - Chapter Summary
		Synchronization	Tannenbaum	Synchronization

Sistem Terdistribusi

8	Fault Tolerance	1. M. V. S. Andrew S. Tanenbaum, Distributed Systems: Principles and Paradigms, 3rd Ed., 2017. 2. R. Vitillo, Understanding Distributed Systems: What every developer should know about large distributed applications, 2nd Ed., 2022. 3. K. Coulouris, Dollimore, Distributed Systems: Concepts and Design 5th edition, 2005	1. Introduction to Fault Tolerance 2. Process Resilience 3. Reliable Client-Server Communication 4. Reliable Group Communication 5. Distributed Commit 6. Failure Detection 7. Common Failure Causes 8. Downstream Resiliency 9. Upstream Resiliency 10. Fault-tolerant Services 11. Recovery 12. Chapter Summary
9	Replika Consistency and Replication	Tannenbaum Tannenbaum	Replika Consistency and replication 1. Introduction to Security 2. Overview of Security Techniques 3. Cryptographic Algorithms 4. Digital Signatures 5. Cryptographic Pragmatics 6. Secure Channels 7. Secure Links 8. Access Control 9. Secure Naming 10. Security Management 11. Manage SELinux Security 12. Chapter Summary
	Security	1. Tanenbaum. Andrew S, Maarten Van Steen, Distributed Systems: Principles and Paradigms, 3rd Ed., Pearson Education, Inc., 2017 (Chapter 9). 2. Coulouris, Dollimore, Kindberg, Distributed Systems: Concepts and Design 5th edition, Addison Wesley: 2005 (Chapter 11.2, 11.3, 11.4, 11.5). 3. Vitillo. Roberto, Understanding Distributed Systems: What every developer should know about large distributed applications, 2nd Ed., 2022 (Chapter 3) 4. Red Hat® Team - RHEL RH134 - EN, Red Hat System Administration II, 2024, https://www.redhat.com/en/services/training/red-hat-academy : 2024 (Chapter 6).	1. Architectures 2. Processes 3. Communications 4. Naming 5. Synchronization 6. Distributed Objects 7. Case Study: CORBA 8. From Objects to Components 9. Case Study: Enterprise Java Beans and Fractal 10. Software as a Service 11. Driving Forces 12. Distributed Transactions
10	Distributed-Object Based Systems and Distributed Transactions	1. Tanenbaum. Andrew S, Maarten Van Steen, Distributed Systems: Principles and Paradigms, 3rd Ed., Pearson Education, Inc., 2017 (Chapter 10). 2. Coulouris, Dollimore, Kindberg, Distributed Systems: Concepts and Design 5th edition, Addison Wesley: 2005 (Chapter 8.2, 8.3, 8.4, 8.5). 3. Velte. Toby, Anthony Velte, Robert Elsenpeter, Cloud Computing, A Practical Approach, McGraw Hill Osborne Media., 2012 (Chapter 9).	Zookeeper introduction Download and configuration. Command line execution. 1. Consistency and Replication 2. Fault Tolerance 3. Security 4. Software plus Services 5. Local Clouds and Thin Clients 6. Functional Decomposition 7. Partitioning 8. Duplication 9. Chapter Summary Zookeeper
	Zookeeper	Zookeeper	
11	Distributed Object-based Systems and Scalability	1. Tanenbaum. Andrew S, Maarten Van Steen, Distributed Systems: Principles and Paradigms, 3rd Ed., Pearson Education, Inc., 2017 (Chapter 11). 2. Velte. Toby, Anthony Velte, Robert Elsenpeter, Cloud Computing, A Practical Approach, McGraw Hill Osborne Media., 2012 (Chapter 10, 12). 3. Vitillo. Roberto, Understanding Distributed Systems: What every developer should know about large distributed applications, 2nd Ed., 2022 (Chapter 12, 13, 14)	
	Zookeeper installation	-	

			1. Architecture 2. File Service Architecture 3. Processes 4. Communication 5. Naming 6. Synchronization 7. Consistency and Replication 8. Fault Tolerance 9. Security 10. Case Study: Sun Network File Systems 11. Case Study: The Andrew File System 12. Distributed Multimedia System 13. Characteristics of Multimedia Data 14. Quality of Services Management 15. Resource Management 16. Stream Adaptation 17. Case Studies: Tiger, BitTorrent and End System Multicast 18. Chapter Summary
	12	Distributed File Systems and Distributed Multimedia Systems	
Sistem Terdistribusi		1. Tanenbaum. Andrew S, Maarten Van Steen, <i>Distributed Systems: Principles and Paradigms</i> , 3rd Ed. © Pearson Education, Inc., 2017 (Chapter 12). 2. Coulouris, Dollimore, Kindberg, <i>Distributed Systems: Concepts and Design</i> 5th edition © Addison Wesley: 2005 (Chapter 12.2, 12.3, 12.4, 20). Tannenbaum	Fault tolerance - 1 1. Architecture 2. Processes 3. Communication 4. Naming 5. Synchronization 6. Consistency and Replication 7. Fault Tolerance 8. Accessing the Cloud 9. Web Applications and APIs 10. Web Services 11. Service Description and IDL for Web Services 12. A Directory Service for Use with Web Services 13. XML Security 14. Coordination of Web Services 15. Applications of Web Services 16. Chapter Summary
	13	Distributed Web-based Systems and Web Services	
		1. Tanenbaum. Andrew S, Maarten Van Steen, <i>Distributed Systems: Principles and Paradigms</i> , 3rd Ed. © Pearson Education, Inc., 2017 (Chapter 13). 2. Coulouris, Dollimore, Kindberg, <i>Distributed Systems: Concepts and Design</i> 5th edition © Addison Wesley: 2005 (Chapter 9). 3. Velte. Toby, Anthony Velte, Robert Elsenpeter, <i>Cloud Computing, A Practical Approach</i> © McGraw Hill Osborne Media., 2012 (Chapter 6). Tannenbaum	Web services 1. Introduction to Coordination Models 2. Architecture 3. Processes 4. Communication 5. Naming 6. Consistency and Replication 7. Fault Tolerance 8. Security 9. Introduction to OpenShift Application 10. Containers 11. Container Orchestration 12. Run Containers 13. Chapter Summary
	14	Distributed Coordination-based Systems and Micro services	
		1. Tanenbaum. Andrew S, Maarten Van Steen, <i>Distributed Systems: Principles and Paradigms</i> , 3rd Ed. © Pearson Education, Inc., 2017 (Chapter 14). 2. Hunter II, Thomas, <i>Distributed Systems with Node.js: Building Enterprise-Ready Backend Services</i> © O'Reilly Media Inc., 2021 (Chapter 1). 3. Red Hat® Team - RHEL RH134 - EN, <i>Red Hat System Administration II</i> © 2024, https://www.redhat.com/en/services/training/red-hat-academy : 2024 (Chapter 12). 4. Red Hat Team, <i>OpenShift 101- Introduction to OpenShift Applications</i> © Red Hat Academy Course, https://rha.ole.redhat.com/rha/app/summary : 2024 (Chapter 5, 7).	Jurnal for UAS 1. Introduction 2. Array, ArrayList, LinkedList
Struktur Data		Kaufmman Gaddis	Jurnal for UAS Penjelasan silabus. Introduction - OOP - Big O Notation - Netbeans Array, ArrayList, LinkedList Exercise Array, Exercise LinkedList

Struktur Data	3	Singly list			Singly Linked List. Node. AppendList. Search element Delete Node
	4	GL	Koffmann	-	-
	5	Stack and queue	Koffman		Stack and queue
	6	Recursion	Kaufmann		Recursion Binary search tree; Penjelasan projek untuk UTS. Review materi untuk UTS
	7	Binary search tree		Kaufmann	Tree-2 Heap tree. MinHeap dan MaxHeap
	8	Tree - 2		Koffmann	Huffman tree Set and map 1 Set, hashTable.
	9	Set and map 1		Koffmann	Hashcode, index HashTable
	10	Map and set naviable	Koffmann		Map and set naviable Sorting 1 Shell sort. Merge sort
	11	Sorting 1		Kofman	Insertion sort Selection sort Tree balancing Right- and left rotation
	12	Tree balancing		Koffmann	Red-Black tree Timsort, Heapsort, Quicksort
	13	Sorting 1		Koffmann	Review and exercise discussion
	14	Review and exercise discussion	Koffmann		1. Introduction 2. OSPF Features and Characteristics 3. OSPF Packets 4. OSPF Operation
Wide Area Network (WAN)	1	Single-Area OSPFv2 Concepts	CCNA version 7.02, Module 3, Enterprise Networking Security and Automation v.7.02 (ENSA) https://contenthub.netacad.com/ensa (Chapter 1)		
	2	Single-Area OSPFv2 Configuration	CCNA version 7.02, Module 3, Enterprise Networking Security and Automation v.7.02 (ENSA) https://contenthub.netacad.com/ensa (Chapter 2)		
	3	Network Security Concepts	CCNA version 7.02, Module 3, Enterprise Networking Security and Automation v.7.02 (ENSA) https://contenthub.netacad.com/ensa (Chapter 3)		
	4	ACL Concepts	CCNA version 7.02, Module 3, Enterprise Networking, Security and Automation v.7.02 (ENSA) https://contenthub.netacad.com/ensa (Chapter 4)		

1. Introduction
2. OSPF Router ID
3. Point-to-Point OSPF Networks
4. Multiaccess OSPF Networks
5. Modify Single-Area OSPFv2
6. Default Route Propagation
7. Verify Single-Area OSPFv2
8. IP Services
9. Network Security Best Practices
10. Cryptography
1. Introduction
2. Current State of Cybersecurity
3. Threat Actors
4. Malware
5. Common Network Attacks
6. IP Vulnerabilities and Threats
7. TCP and UDP Vulnerabilities
8. IP Services
9. Network Security Best Practices
10. Cryptography
1. Introduction
2. Purpose of ACLs
3. Wildcard Masks in ACLs
4. Guidelines for ACL Creation
5. Types of IPv4 ACL
1. Introduction
2. Purpose of ACLs
3. Wildcard Masks in ACLs
4. Guidelines for ACL Creation
5. Types of

Wide Area Network (WAN)

5	ACLs for IPv4 Configuration	CCNA version 7.02, Module 3, Enterprise Networking, Security and Automation v.7.02 (ENSA) (Chapter 5)	1. Introduction 2. Configure Standard IPv4 ACLs 3. Modify IPv4 ACLs 4. Secure VTY Ports with a Standard IPv4 ACL 5. Configure Extended IPv4 ACLs
6	NAT for IPv4	CCNA version 7.02, Module 3, Enterprise Networking Security and Automation v.7.02 (ENSA) (Chapter 6)	1. Introduction 2. Configure Standard IPv4 ACLs 3. Modify IPv4 ACLs 4. Secure VTY Ports with a 5. Configure Extended IPv4 ACLs 1. Introduction 2. NAT Characteristics 3. Types of NAT 4. NAT Advantages and Disadvantages 5. Static NAT 6. Dynamic NAT 7. PAT 8. NAT64
7	WAN Concepts and Quiz before Mid Term Examination	CCNA version 7.02, Module 3, Enterprise Networking, Security and Automation v.7.02 (ENSA) (Chapter 7)	1. Introduction 2. Purpose of WANs 3. WAN Operations 4. Traditional WAN Connectivity 5. Modern WAN Connectivity 6. Internet-based Connectivity 1. Introduction 2. Purpose of WANs 3. WAN Operations 4. Traditional WAN Connectivity 5. Modern
8	VPN and IPSec Concepts	CCNA version 7.02, Module 3, Enterprise Networking, Security and Automation v.7.02 (ENSA) (Chapter 8)	1. Introduction 2. VPN Technology 3. Types of VPNs 4. IPSec
9	QoS Concepts	CCNA version 7.02, Module 3, Enterprise Networking, Security and Automation v.7.02 (ENSA) (Chapter 9)	1. Introduction 2. Network Transmission Quality 3. Traffic Characteristics 4. Queueing Algorithms 5. QoS Models 6. QoS Implementation Techniques 1. Introduction 2. Network Transmission Quality 3. Traffic Characteristics 4. Queueing Algorithms 5. QoS
10	Network Management	CCNA version 7.02, Module 3, Enterprise Networking, Security and Automation v.7.02 (ENSA) (Chapter 10)	1. Introduction 2. Device Discovery with CDP 3. Device Discovery with LLDP 4. NTP 5. SNMP 6. Syslog 7. Router and Switch File Maintenance 8. IOS image management 1. Introduction 2. Device Discovery with CDP 3. Device Discovery with LLDP 4. NTP 5. SNMP 6. Syslog 7. Router and Switch File Maintenance 8. IOS image management
11	Network Design	CCNA version 7.02, Module 3, Enterprise Networking, Security and Automation v.7.02 (ENSA) (Chapter 11)	1. Introduction 2. Hierarchical Networks 3. Scalable Networks 4. Switch Hardware 5. Router Hardware
12	Network Troubleshooting	CCNA version 7.02, Module 3, Enterprise Networking Security and Automation v.7.02 (ENSA) (Chapter 12)	1. Introduction 2. Network Documentation 3. Troubleshooting Process 4. Troubleshooting Tools 5. Symptoms and Causes of Network Problems 6. Troubleshooting IP Connectivity
		CCNA version 7.02, Module 3, Enterprise Networking Security and Automation v.7.02 (ENSA) (Chapter 1)	1. Introduction 2. Network Documentation 3. Troubleshooting Process 4. Troubleshooting Tools 5. Symptoms and Causes of Network Problems 6. Troubleshooting IP Connectivity

Wide Area Network (WAN)	13	Network Virtualization	CCNA version 7.02, Module 3, Enterprise Networking, Security and Automation v.7.02 (ENSA) https://contenthub.netacad.com/ensa (Chapter 13)	1. Introduction 2. Cloud Computing 3. Virtualization 4. Virtual Network Infrastructure 5. Software-Defined Networking 6. Controllers 1. Introduction2. Cloud Computing3. Virtualization4. Virtual Network Infrastructure5. Software-Defined Networking6. Configuration Management Tools7. IBN and Cisco DNA Center
	14	Network Automation		
	Grand Total			